

Decomposition Approach to the Public Transport Scheduling Problem*

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Abstract—This paper presents a new approach for solving timetable optimization problems. The original problem is decomposed into subproblems along the discrete time axis. For each subproblem the motion of vehicles is described by treating headways between them as the state variables. The layover times at the terminals are considered as the decision variables. This model allows the implementation of optimal control theory with quadratic criterion and technical constraints imposed on the state and control. The problem is to obtain after a fixed number of trips a desired configuration of headways and to “glue” partial timetables together creating the final timetable. A numerical example is given to illustrate the idea.

1. Introduction

THE PRECEDING DECADE has seen a considerable increase in the number and variety of computer assisted techniques being applied to the bus scheduling process (Wren, 1980; Williamson, 1985; Rousseau and Blais, 1985). These techniques have concentrated on two approaches. The first group of methods tried to capture the manual process in a computer code. A good example of this approach was given by Iida (1988), who described the application of an artificial intelligence method to timetable preparation of a single-track railway.

An interesting scheme in handling modeling and timetable preparation problems for fixed route systems originated from the discovery that the vehicle movements can be described by a state-space model, making possible the implementation of optimal control methods (Curry *et al.*, 1980; Araya and Sone, 1984). This method was adopted by Breusegem *et al.* (1988) in order to generate the optimal sequence of the departure times of metro trains.

The essence of these models can be stated quite simply. For a fixed route system bus movements are viewed as a discrete events process. An event corresponds to bus departure from the terminal. The discrete equations, obviously called “interval equation”, describe the evolution of headways for successive trips of vehicles. The layover times allowed at the terminal points are decision variables. The objective function consists of three terms. The first term penalizes the deviations of the time intervals (“headways”) between vehicles and is therefore related to the average waiting time for the passengers. The second term penalizes the deviations of control variables from the nominal values. The last term of the objective function estimates a deviation of the final state from its desired value. The problem is to determine the optimal sequence of the layover times at the

terminal that minimizes the objective function subject to the interval equations and constraints.

For many optimal control problems, the long time horizon of planning adds another dimension of difficulty. This is the case for the model considered in this paper. The service time is divided into subperiods. Within each subperiod the number of buses is fixed, according to the travel demand and available resources. The fixed number of buses determines the headway at the beginning of each subperiod, but not the assignment of buses to the task. The assignment of buses to the tasks determines the final configuration of headways. The structure of the problem can be explored to seek for efficient solution methodology. The final state of “ m ” subperiod, is also the initial state of “ $m + 1$ ” subperiod. Define subproblem “ m ” as the optimization problem over the subperiod “ m ”, having the corresponding objective function and given initial and final states. Inside the subperiod the optimal sequence of the departure times can be obtained (Grega, 1989) creating the partial timetable.

The following conditions should be fulfilled in order to make possible the connection of the partial timetables:

- (1) an optimal assignment of buses to the tasks must be decided for each subperiod, in the form of desired final states;
- (2) a desired configuration of headways must be reached after a fixed number of trips, as the result of optimization of the partial timetable. The problem then reduces to how to design an algorithm to search for optimal configuration of the desired final states. To do this, an upper level optimization problem is created.

To illustrate the idea, let us consider the preparation of a timetable covering three time periods, each characterized by a given number of vehicles, the number of trips and the initial and final states (Fig. 1). The desired final states $\bar{x}(K^1)$, $\bar{x}(K^2)$, $\bar{x}(K^3)$ are determined by an upper-level algorithm, subject to some bus assignment constraints. It can be easily verified that the set of desired final states is finite and consists of two solutions after the first subperiod, and six solutions after the second subperiod. The timetable preparation problem is solved sequentially and the partial timetables are glued together creating the final timetable. The first partial timetable in the sequence is generated with the given initial state $x^1(0)$ and desired final state $\bar{x}(K^1)$. The state $x(K^1)$ attained after optimization of the partial timetable 2 is used as the initial state for optimization of the next partial timetable.

The paper is organized as follows. First, a state-space model of a bus route is described and the objective function is introduced. Then a two-level optimization problem is proposed taking into account the structure of the problem. Afterwards the solution algorithms are proposed for the lower and upper level, respectively. Finally, an example is given illustrating the method as it applies to a simple situation.

2. The model

2.1. Terminology and basic notations. We introduce here the basic notations for the model and some related definitions which characterize the problem.

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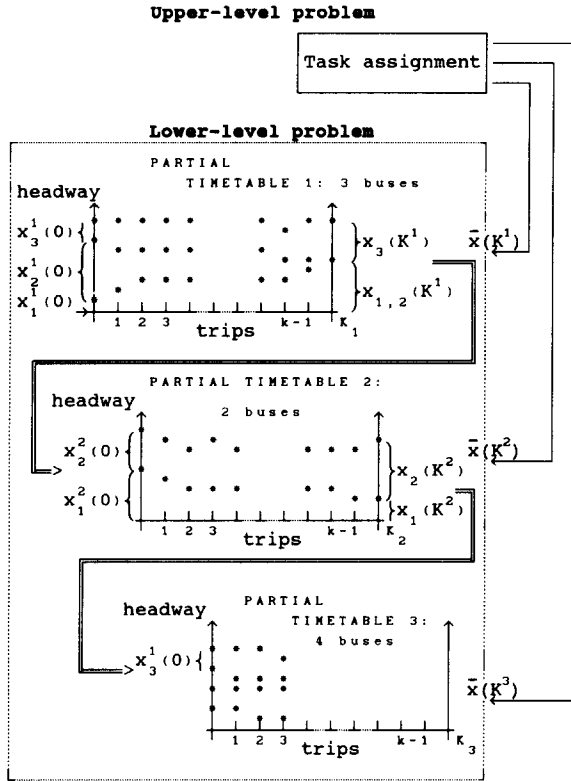


FIG. 1. Generation of partial timetables.

Buses operate along a fixed route. A bus route consists of two terminals A and B and intermediate stops over which service is provided. The service time is divided into $M + 1$ subperiods. These need not be of equal duration.

The passenger demand is estimated and translated into the number of buses needed during every subperiod. Within each subperiod the number of buses is fixed and determines the level of the service in the form of the tasks to be done.

Bus working patterns are given for the service time. The goal here is to make the bus work continuously as long as possible, minimizing empty running time.

At the beginning of each subperiod the assignment of buses to the tasks should be decided. The tasks are arranged according to assumed bus working patterns to determine the sequences of bus operations.

A single two-way traversal of the route is called a trip. It consists of two runs between terminal points and layover time at the terminal A. Trip times may differ over the subperiod due to the different layover times. Assigning a layover at the terminal covers three types of function: drivers resting time, departure reliability, and control variable when optimizing a timetable. The last one of these is considered in this paper.

A route timetable for m subperiod is defined to be a sequence of trips assigned to the buses:

$$\{t_i(k)\}, \quad i = 1 \cdots N^m, \quad k = 1 \cdots K^m,$$

where $t_i(k)$ is the departure time from the terminal A of i bus during k -trip.

The basic assumptions in the model are the following.

(1) Generation of timetables is broken down into stages to reduce the complexity of the process. Two stages of the process are described by the model:

- assignment of buses to the tasks at the beginning of each subperiod;
- assignment of departure times to the buses within the subperiod.

(2) Overtaking of buses is not allowed within the subperiod.

(3) Upper and lower bounds on layover time are given for each subperiod.

2.2. State equations. It is possible to describe the motion of buses by treating headways between them as the state variables and trips as the stages of a discrete process. A discrete model of the timetable generation process during m subperiod is given by the set of difference equations:

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{I}^m \mathbf{x}(k) + \mathbf{B}^m \mathbf{u}(k) + \mathbf{B}_1^m \mathbf{u}(k-1), \\ \mathbf{x}(k), \mathbf{u}(k), \mathbf{u}(k-1) &\in R^{N^m}, \end{aligned} \quad (1)$$

where $x(k)$ is a headway defined as:

$$\begin{aligned} x_i(k) &= t_i(k) - t_{i-1}(k), \quad i = 2 \cdots N, \\ x_1(k) &= t_1(k) - t_N(k-1). \end{aligned} \quad (2)$$

$\mathbf{I}^m$ is $N^m \times N^m$ identity matrix and $\mathbf{u}(k)$ is the layover time at terminal A. The variable $\mathbf{u}(k)$ will be seen as a control variable. $\mathbf{B}^m, \mathbf{B}_1^m$ are $N^m \times N^m$ matrices in the form:

$$\begin{aligned} \mathbf{B}^m &= \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1 & 0 & 0 & \cdots & 0 & 0 \\ & 0 & 0 & 0 & \cdots & -1 & 1 \end{bmatrix}, \\ \mathbf{B}_1^m &= \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots & 0 & -1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}, \end{aligned} \quad (3)$$

representing interconnections between adjacent buses in the string. For $k = 0$, we get the initial conditions:

$$\begin{aligned} x_i^m(0) &= t_i^m(0) - t_{i-1}^m(0), \\ x_1^m(0) &= t_1^m(0) - t_N^m(-1), \\ i &= 2 \cdots N^m. \end{aligned}$$

For $m = 1$, $t_N^1(-1)$ is the opening time of the service and:

$$u_i(-1) = u_{i0} = 0, \quad \sum_{i=1}^N x_i^m(0) = T^m,$$

where T^m is the route travel time during m subperiod.

2.3. Transformation of the state variables. The timetable generating procedure can be seen as a long horizon control problem decomposed into smaller subproblems along a discrete time axis. Subproblem number (m) starts from the initial state $\mathbf{x}^m(0)$. The final state $\mathbf{x}(K^m)$ can be used as the initial state for $(m+1)$ subproblem. Since $\mathbf{x}^1(0)$ is given initially, the procedure is well-defined when the final states are determined uniquely.

From m subproblem a suitable transformation of headways can be defined for pair $\{\mathbf{x}^m(0), \mathbf{x}^m(K^m)\}$:

$$\mathbf{x}(K^m) = \mathbf{S}^m \mathbf{x}^m(0). \quad (4)$$

The matrix $\mathbf{S}^m$ is the diagonal matrix $N^m \times N^m$ consisting of headway scaling coefficients.

It should be noted that the dimension of the state vector $\mathbf{x}^m(k)$ depends on the index of the subproblem, in general $N^m \neq N^{m+1}$. Therefore, suitable transformation formulas can be introduced:

$$\begin{aligned} \mathbf{x}^{m+1}(0) &= \mathbf{P}^m \mathbf{x}(K^m), \\ \mathbf{x}(K^m) \in R^{N^m} \quad \text{and} \quad \mathbf{x}^{m+1}(0) \in R^{N^{m+1}}, \quad N^m \neq N^{m+1}. \end{aligned}$$

The matrix $\mathbf{P}^m$ is $N^{m+1} \times N^m$ matrix with the structure:

$$P_{ij}^m = \begin{cases} 1 & \text{if the bus number } j, \text{ operating during the} \\ & \text{subperiod } m \text{ is assigned to the task} \\ & \text{number } i \text{ during, the subperiod } m+1, \\ 0 & \text{otherwise.} \end{cases}$$

Figure 2 illustrates the idea of this approach.

For $N^m < N^{m+1}$ the transformation equation takes the form:

$$\mathbf{x}^{m+1}(0) = (\bar{\mathbf{P}}^m + \bar{\mathbf{P}}^m) \mathbf{x}(K^m),$$

where: $\bar{\mathbf{P}}^m$ is the task assignment matrix for buses continuing the service, $\bar{\mathbf{P}}^m$ is the matrix of new buses entered at the beginning of $(m+1)$ subperiod (Fig. 3).

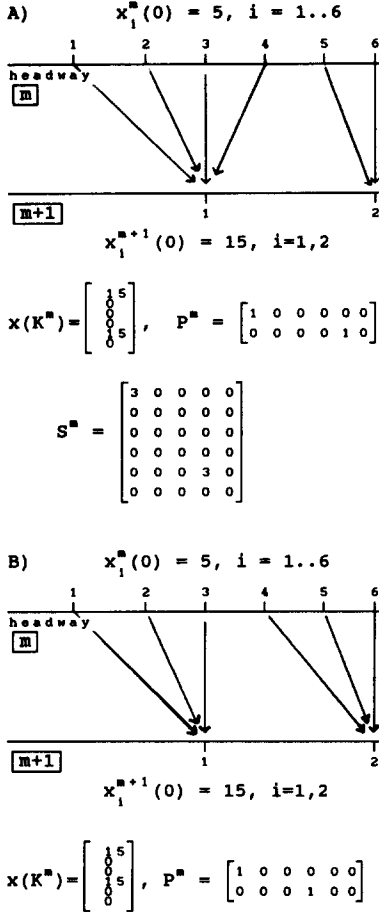


FIG. 2. State transformation matrix: an example for decreasing number of vehicles and two choices of the final states.

The following transformed initial state vector can be defined for $(M+1)$ subperiod:

$$x^{M+1}(0) = \Pi^M S^M \Pi^{M-1} S^{M-1} \dots \Pi^1 S^1 \dots \Pi^m S^m x^m(0), \quad (5)$$

where: $\Pi^i = \bar{\Pi}^i$ for increasing number of vehicles,
 $\Pi^i = P^i$ for decreasing number of vehicles.

Using these definitions the following corollaries can be established:

- (1) a triple $\{x(K^m), P^m, S^m\}$ provides the tasks assignment to the buses at the beginning of $(m+1)$ subperiod;
- (2) non-zero elements of $x_c^{M+1}(0)$ correspond to the buses operating continuously during the subperiods $m, m+1, \dots, M$.

2.4. *Objective function.* For the m subperiod the objective function is assumed in the form:

$$\mathcal{J}^m(x(K^m), x, u) = \sum_{k=0}^{K^m-1} \mathcal{J}^m(x(k), u(k), k) + \mathcal{J}^m(x(K^m)), \quad (6)$$

where:

$$\mathcal{J}^m(x(k), u(k), k) = \frac{1}{2} \{ \|x(k) - \bar{x}^m(k)\|_{R^m(k)}^2 + \|u(k) - \bar{u}^m(k)\|_{Q^m}^2 \}. \quad (7)$$

$\mathcal{J}^m(x(K^m))$ is the terminal cost:

$$\mathcal{J}^m(x(K^m)) = \frac{1}{2} \|x(K^m) - \bar{x}(K^m)\|_{R(K^m)}^2,$$

where:

$$x = (x(0)^T, x(1)^T, \dots, x(K^m-1)^T)^T, \\ u = (u(0)^T, u(1)^T, \dots, u(K^m-1)^T)^T.$$

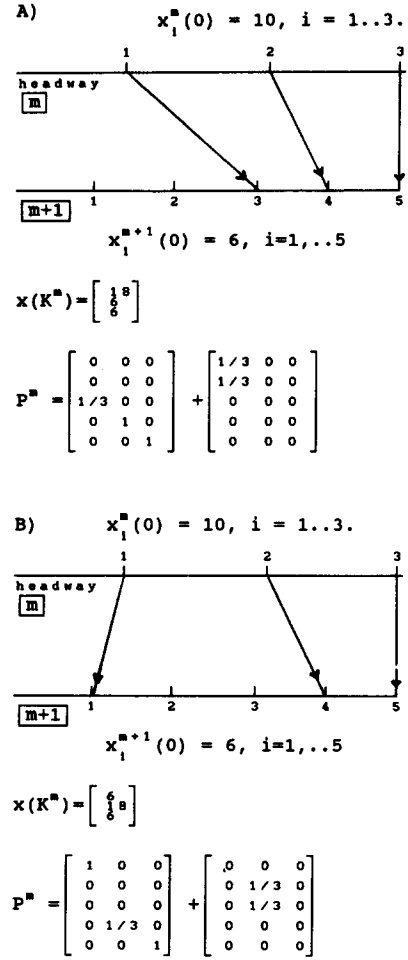


FIG. 3. State transformation matrix: an example for increasing number of vehicles and two choices of the final states. New buses are assigned to the tasks (A) 1, 2; (B) 2, 3.

The first term of the objective function represents the average delay for passengers waiting for a group of N buses. It penalizes the deviations of the states from their desired values $\bar{x}$.

Matrix R describes the travel demand observed along the route. The elements $r_i(k)$ correspond to the statistical mean values of passenger per hour which enter the stops. The matrix Q is diagonal and positively definite.

The state and control variables are constrained by upper and lower bounds:

$$0 \leq x(k) \leq x(k)^{\max}, \quad u^{\min}(k) \leq u(k) \leq u(k)^{\max}. \quad (8)$$

2.5. *Optimization problem.* For m subperiod, the problem of determining the optimal timetable can be considered as the parameter optimization problem:

$$\min_u \mathcal{J}^m(x(K^m), x, u), \quad (9)$$

subject to system dynamics (1) and constraints (8). Since the solution of (9) depends upon $x^m(0)$ and $x(K^m)$, the set of parameters $\{x(K^m), m=1 \dots M\}$ can be chosen so that the total cost for subproblems $1 \dots M$ is minimized. That is:

$$\min_{x(K^m) \in X^m} \sum_{m=1}^M \min_u \mathcal{J}^m(x(K^m), x, u), \quad (10)$$

subject to (1), (8) and service continuity constraint (5), where $x^{M+1}(0)$ is a vector of desired non-zero initial states for the last subperiod. The condition (5) guarantees that buses assigned to the tasks during the subperiod number (m) can also be assigned during next subperiods $(m+1, \dots, M)$ (continuing buses).

3. Solution method

3.1. *Two-level optimization.* The problem (10), is decomposed into M subproblems. Each subproblem consists in optimizing the subperiod timetables on the basis of fixed final costs $\mathbf{x}(K^m)$, taking into account local constraints. It is assumed that for all $\mathbf{x}(K^m) \in \mathbb{X}^m$ each subproblem has an optimal solution: there exists a feasible sequence of controls that brings the system from $\mathbf{x}^m(0)$ to $\mathbf{x}(K^m)$.

The solution of the upper-level problem selects the best sequence $\{\mathbf{x}(K^1), \mathbf{x}(K^2), \dots, \mathbf{x}(K^M)\}$, $\mathbf{x}(K^m) \in \mathbb{X}^m$, so that the overall cost for subproblems $1 \dots M$ is minimized.

3.2. *The solution method for lower-level subproblems.* The main requirement of an optimization method for a lower-level subproblem is the ability to handle a system of high dimensionality with transport delays, and inequality constraints on states and controls. Traditionally in such problems the space is augmented by introducing additional variables for the delay terms and by converting the original high-order difference equation to a set of coupled first order equations. However, this increases the dimensionality and consequently makes the computational burden even more severe. To overcome this difficulty a hierarchical approach due to Lasdon (1970), Tamura (1975), Singh *et al.* (1975) and Papageorgiu (1983) was adapted. For the interested reader, details are given by Grega (1989). The method is based on the observation that the dual of the original problem can be decomposed (by the index k) into K independent small minimization problems which are solved analytically.

3.3. *Upper level optimization.* The solution of the upper-level problem selects the best sequence $\{\mathbf{x}(K^1), \mathbf{x}(K^2), \dots, \mathbf{x}(K^M)\}$, so that the overall cost is minimized. The sets $\mathbb{X}^m$ of the final states are finite for each subproblem and can be calculated as the all possible assignments of buses to the tasks according to assumed bus working patterns.

This problem can be phrased as a directed graph, where the set of nodes contains one node for each feasible final state and $(M - 1)$ nodes for initial states (Fig. 4). Two groups of arcs are created for each subperiod of the workday:

- arcs from $\mathbf{x}^m(0)$ to $\mathbf{x}(K^m)$ corresponding to the solutions of lower level problems;
- arcs from $\mathbf{x}(K^m)$ to $\mathbf{x}^{m+1}(0)$ corresponding to the transformation of state variables.

The feasible solution of the upper level problem corresponds in the graph to a path from node $\mathbf{x}^1(0)$ to node $\mathbf{x}^{M+1}(0)$ satisfying the service continuity constraint (5).

The optimal upper level solution corresponds to a feasible path, which minimizes the sum of the lower level optimal

objectives values:

$$\min_{\mathbf{x}(K^m) \in \mathbb{X}^m} \sum_{m=1}^M \mathcal{J}^m(\mathbf{x}(K^m), \mathbf{x}^*, \mathbf{u}^*), \text{ subject to (5).}$$

A shortest path method can be used to solve this problem optimally. The chosen solution strategy is to use the service continuity constraints (5) in order to eliminate nonfeasible solutions. The steps of the method are as follows:

- (1) Build the feasible path in network (Fig. 4) using the following rule for the (m) arc:

$$J^{m*} = \min_{\mathbf{x}(K^m) \in \mathbb{X}^m} \mathcal{J}(\mathbf{x}(K^m), \mathbf{x}^*, \mathbf{u}^*) + J^{m+1*} + \dots + J^{M*},$$

subject to:

$$\mathbf{x}^{M+1}(0) = \Pi^{M*} \mathbf{S}^{M*} \Pi^{M-1*} \dots \Pi^{1*} \mathbf{S}^{1*} \dots \Pi^m \mathbf{S}^m \mathbf{x}^m(0),$$

where Π^{i*} , $\mathbf{S}^{i*}$ are transformation matrices corresponding to $\mathbf{x}(K^i)$ solution. Set $\bar{J} = J^{m*}$ as the upper bound on the value of the optimal solution.

- (2) Starting from $\mathbf{x}^{M+1}(0)$ consider the other feasible paths in the network:

- if $J^{m*} < \bar{J}$ then $\bar{J} = J^{m*}$,
- if at the node of the network the solution is nonfeasible then eliminate this path.

- (3) Terminate if all paths were explored according to the above strategy, $\bar{J}$ is the solution.

4. Test results

In order to verify the above described algorithms a typical four-period timetable preparation problem was considered. The service time was divided into four subperiods ($M = 3$). The following pattern of service was assumed (Table 1).

The initial state for the first subperiod was: $x_1^1(0) = 10$, $x_2^1(0) = 10$, $x_3^1(0) = 10$. The desired initial state for $M + 1$ subperiod was: $\mathbf{x}^4(0) = (15, 15)$. The parameters of the model were: number of trips $K^m = 10$, $u_i^{\max} = 2$, $u_i^{\min} = 0$. For the lower level optimization the performance weighting matrices $\mathbf{Q}$ and $\mathbf{R}$ were taken to be diagonal with the values $q_i = 1$, $r_i(k) = 10$ for $k = 0, \dots, 9$ and $r_i(10) = 20$.

The values of the dual variable p were initialized at zero. The iterations for subproblems were stopped when the absolute values of the gradient were less than one. The convergence was achieved after 25–30 iterations.

The assignment of buses to the tasks, generated by the upper level optimization algorithm in the form of the optimal final states, is given in Table 2. The total cost, given by (10) is the smallest possible for the solution given in Table 2.

Figure 5 shows the evolution of the headways (in minutes) between successive buses. For each subperiod and each bus the optimal control sequences (layover times) are visualized, as well. The interpretation of these computational results is reasonable. The gradual movement of headways towards their desired final values can be observed, giving the smoothing effect, while the regularity of service is kept in the first part of each subperiod. The partial timetables can be “glued” together creating the final timetable.

TABLE 1. PATTERN OF SERVICE

Subperiod no:	1	2	3	4
Number of buses	3	5	6	2

TABLE 2. ASSIGNMENT OF BUSES TO THE TASKS

Subperiod	Optimal final state	Bus no:	Assignment of buses to the tasks					
			1	2	3	4	5	6
1	(12, 6, 12)	Tasks	1	2	3			
2	(5, 10, 5, 5, 5)		2	3	5	1	4	
3	(0, 0, 15, 0, 0, 15)		3	4	6	1	5	2

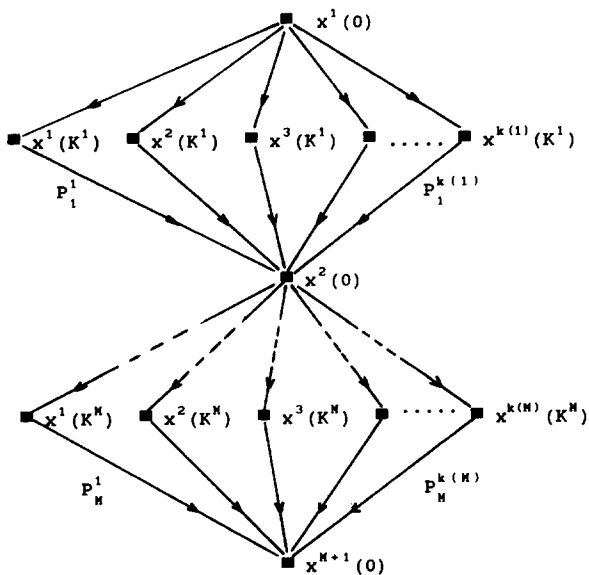


FIG. 4. Initial and final states branching tree.

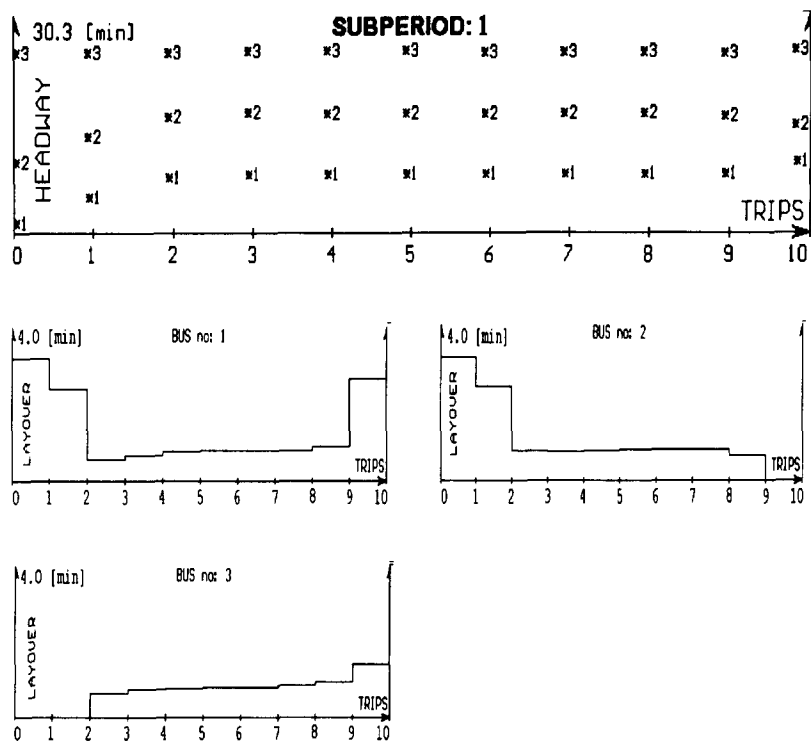


FIG. 5(a). Results of the numerical test, subperiod 1.

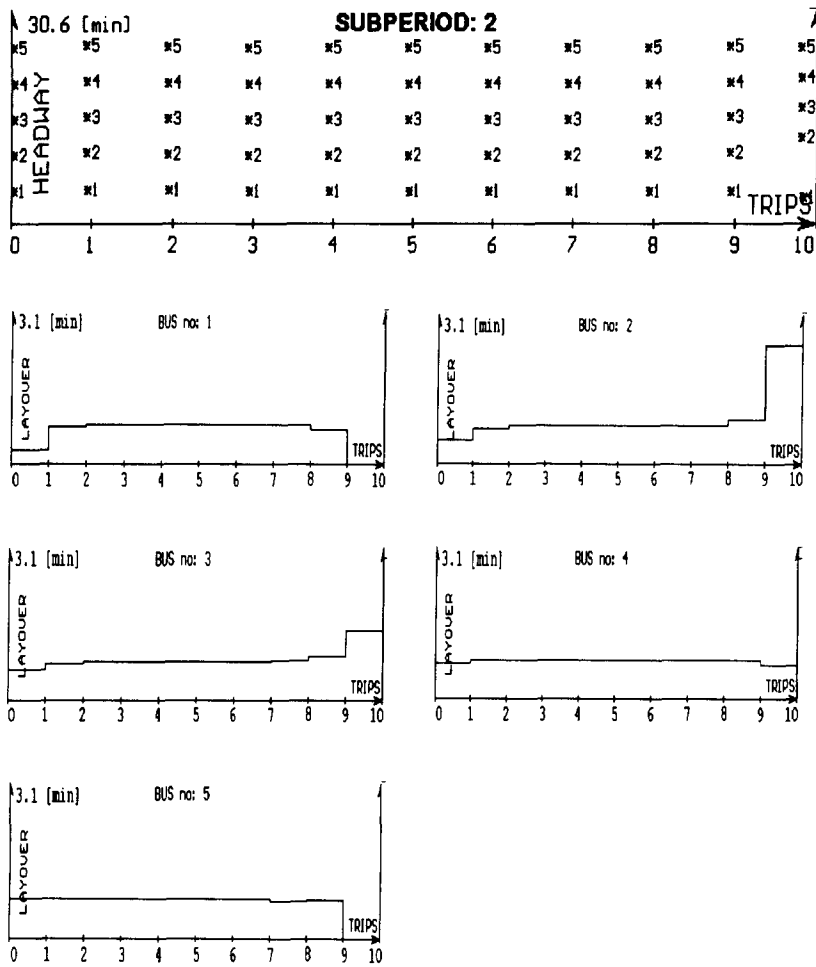


FIG. 5(b). Results of the numerical test, subperiod 2.

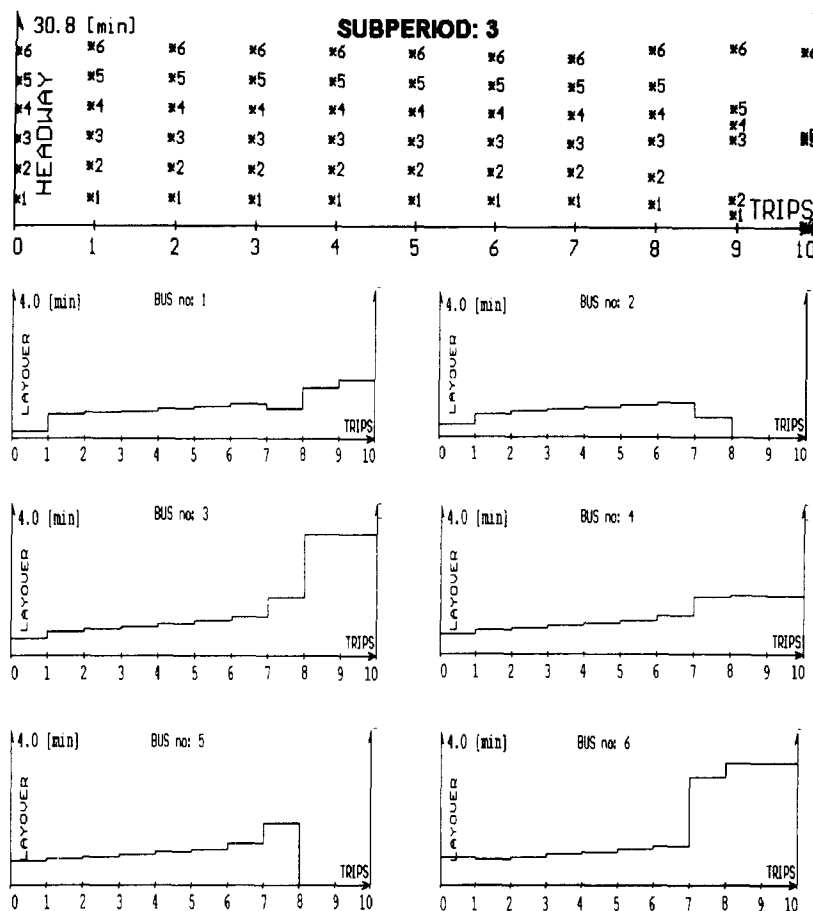


FIG. 5(c). Results of the numerical test, subperiod 3.

5. Conclusions

The urban fixed route system was considered. Following the study on application of optimal control methods in handling timetable preparation problems (Grega, 1989) a two-level approach was proposed. The problem was decomposed on the basis of its structure. The idea was to divide a multistage problem into subproblems along the discrete time axis. The evolution of headways for successive trips was described by the set of discrete equations. The appropriate objective function was introduced. The requirement that for the continuing buses the initial state of a subproblem equals to the final state of the preceding subproblem was formulated in the form of the constraints. The upper-level optimization algorithm was introduced in order to select the best set of the final states. It can be concluded that the fundamental idea was correct and an optimal timetable can be generated with a computer within a reasonable processing time. The methodology described in the paper was implemented in timetable compiling sections of some Polish bus transport organizations.

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